

Project - 2: Website Orchestration

Industry: Internet related

Problem Statement:

How to orchestrate a website with lesser time and higher availability along with Auto Scaling.

Topics:

In this AWS project, you have to deploy a high-availability PHP application with an external Amazon RDS database to Elastic Beanstalk. Running a DB instance external to Elastic Beanstalk decouples the database from the lifecycle of your environment. This lets you connect to the same database from multiple environments, swap one database for another, or perform a blue/green deployment without affecting your database.

Highlights:

Launch a DB instance in Amazon RDS

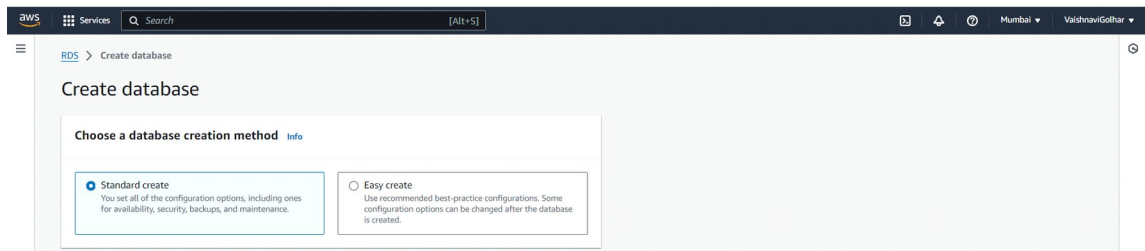
Create an Elastic Beanstalk Environment

Configure Security and scaling

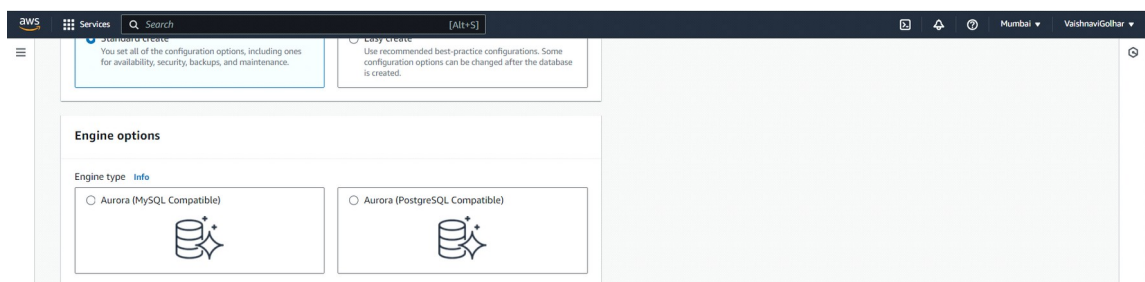
Solution:

1. Launch a DB instance in Amazon RDS

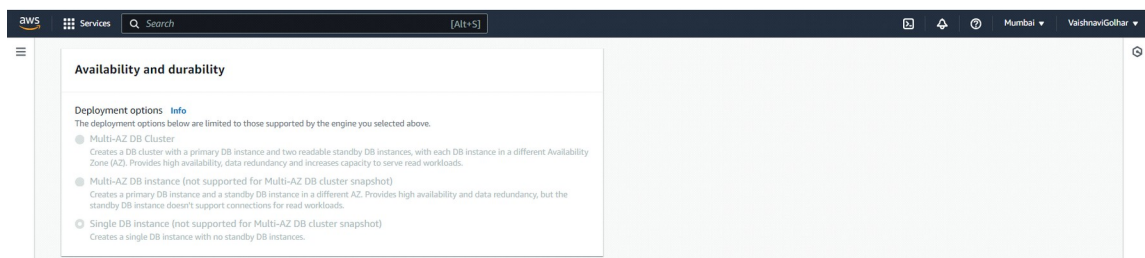
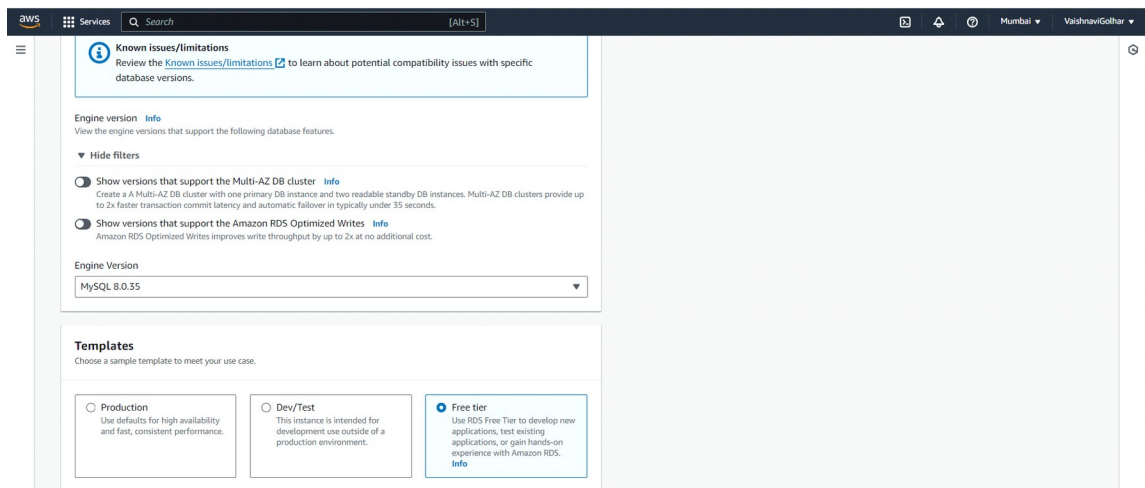
1. To launch an RDS DB instance in a default VPC
2. Open the RDS console. In the navigation pane, choose Databases. Choose Create database. Choose Standard Create.



3. select the engine type (e.g. MySQL).



4. Choose the appropriate instance specifications, including instance class, storage, and allocated storage.



5. Configure the database settings such as DB instance identifier, master username, and password.

The screenshot shows the 'Settings' page in the AWS Management Console. The 'DB instance identifier' field is set to 'ebdbinstance'. Under 'Credentials Settings', the 'Master username' is 'admin'. The 'Credentials management' section has 'Self managed' selected. The 'Master password' field is masked with asterisks, and the 'Confirm master password' field is also masked. The page includes various informational links and constraints for each field.

6. Configure additional settings like VPC, subnet group, security group, backup retention, etc.

The first screenshot shows the 'Instance configuration' page. The 'DB instance class' is set to 'db.t3.micro'. The 'Show instance classes that support Amazon RDS Optimized Writes' checkbox is checked. The 'Include previous generation classes' checkbox is also checked. The 'Standard classes (includes m classes)' checkbox is checked. The 'Memory optimized classes (includes r and x classes)' checkbox is checked. The 'Burstable classes (includes t classes)' checkbox is checked. The 'db.t3.micro' class is selected in the dropdown menu.

The second screenshot shows the 'Storage' page. The 'Storage type' is set to 'General Purpose SSD (gp2)'. The 'Allocated storage' is set to '20' GiB. A notification box states: 'After you modify the storage for a DB instance, the status of the DB instance will be in storage-optimization. Your instance will remain available as the storage-optimization operation completes. Learn more'. The 'Storage autoscaling' section is expanded, showing the 'Enable storage autoscaling' checkbox, which is currently unchecked.

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DB subnet group Info

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default

Public access Info

☒ Yes

RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☐ No

RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (Firewall) Info

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ Choose existing

Choose existing VPC security groups

☐ Create new

Create new VPC security group

Existing VPC security groups

Choose one or more options

RDS database sg X

Availability Zone Info

No preference

RDS Proxy

Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

☐ Create an RDS Proxy Info

RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see Amazon RDS Proxy pricing.

Certificate authority - optional Info

Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

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so by checking the server certificate that is automatically installed on all databases that you provision.

rds-ca-rsa2048-g1 (default)
Expiry: May 20, 2061

If you don't select a certificate authority, RDS chooses one for you.

Additional configuration

Database port info
TCP/IP port that the database will use for application connections.
3306

Tags

A tag consists of a case-sensitive key-value pair.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Database authentication

Database authentication options info

- Password authentication
Authenticates using database passwords.
- Password and IAM database authentication
Authenticates using the database password and user credentials through AWS IAM users and roles.
- Password and Kerberos authentication
Choose a directory in which you want to allow authorized users to authenticate with this DB instance using Kerberos Authentication.

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Enable auto minor version upgrade

Enabling auto minor version upgrade will automatically upgrade to new minor versions as they are released. The automatic upgrades occur during the maintenance window for the database.

Maintenance window

info

Select the period you want pending modifications or maintenance applied to the database by Amazon RDS.

Choose a window

No preference

Deletion protection

Enable deletion protection

Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.

Estimated monthly costs

The Amazon RDS Free Tier is available to you for 12 months. Each calendar month, the free tier will allow you to use the Amazon RDS resources listed below for free:

750 hrs of Amazon RDS in a Single-AZ db.t2.micro, db.t3.micro or db.t4g.micro Instance.

20 GB of General Purpose Storage (SSD).

20 GB for automated backup storage and any user-initiated DB Snapshots.

Learn more about AWS Free Tier.

When your free usage expires or if your application use exceeds the free usage tiers, you simply pay standard, pay-as-you-go service rates as described in the Amazon RDS Pricing page.

You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.

Cancel

Create database

7. Review the configuration and click on "Create database" to launch the RDS instance.

Next, modify the security group attached to your DB instance to allow inbound traffic on the appropriate port. This is the same security group that you will attach to your Elastic Beanstalk environment later, so the rule that you add will grant ingress permission to other resources in the same security group.

8. To modify the inbound rules on the security group that's attached to your RDS instance.

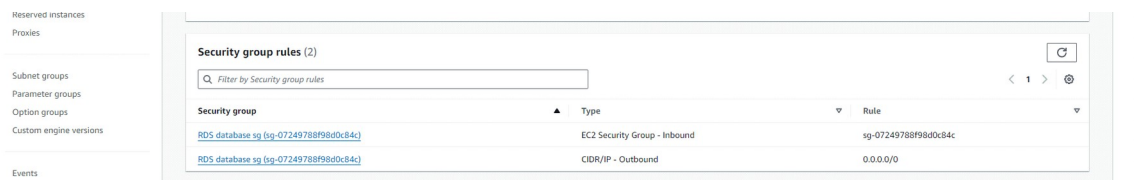
A. Open the Amazon RDS console. Choose Databases. Choose the name of your DB instance to view its details.

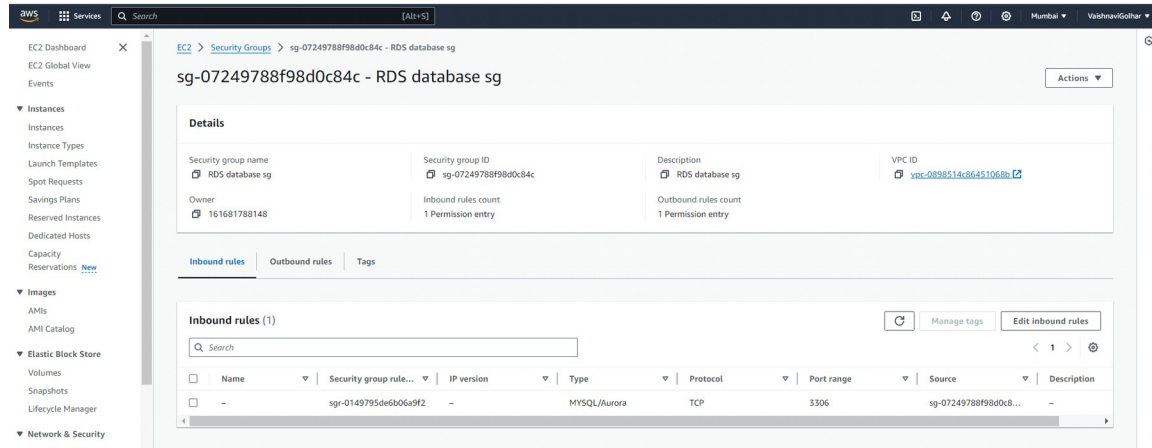
In the Connectivity section, make a note of the Subnets, Security groups, and Endpoint that are displayed on this page. This is so you can use this information later.

B. Under Security, you can see the security group that's associated with the DB instance. Open the link to view the security group in the Amazon EC2 console.

In the security group details, choose Inbound. Choose Edit. Choose Add Rule. For Type, choose the DB engine that your application uses.

For Source, type sg- to view a list of available security groups. Choose the security group that's associated with the Auto Scaling group that's used with your Elastic Beanstalk environment. This is so that Amazon EC2 instances in the environment can have access to the database.





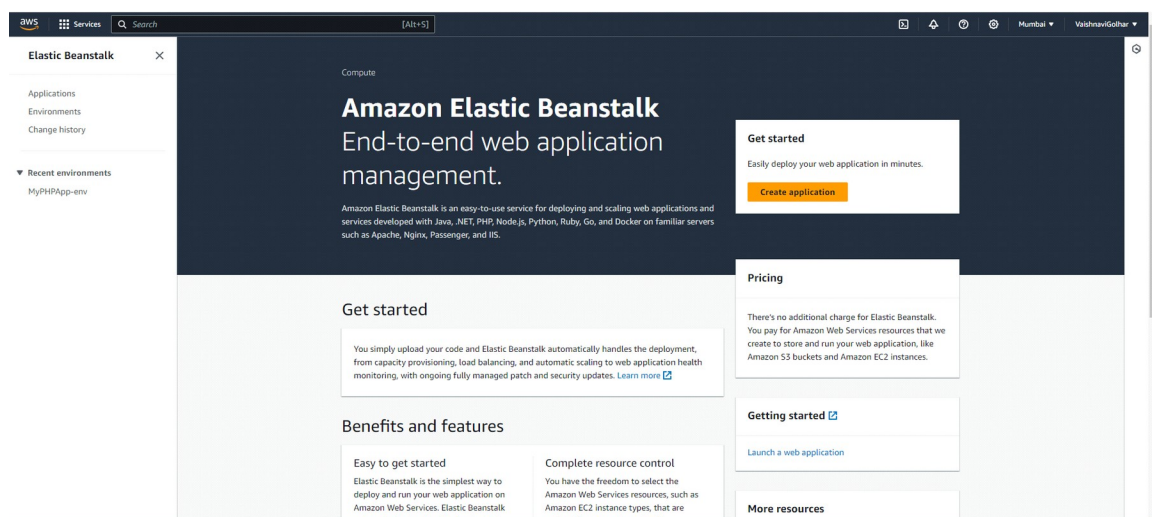
Creating a DB instance takes about 10 minutes. In the meantime, create your Elastic Beanstalk environment.

2. Create an Elastic Beanstalk environment

1. Open the Elastic Beanstalk console. For Platform, select the platform and platform branch that match the language used by your application.
2. For Application code, choose Sample application.

Choose Review and launch.

Review the available options. Choose the available option you want to use, and when you're ready, choose Create app.



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Step 1
Configure environment

Step 2
Configure service access

Step 3 - optional
Set up networking, database, and tags

Step 4 - optional
Configure instance traffic and scaling

Step 5 - optional
Configure updates, monitoring, and logging

Step 6
Review

Configure environment info

Environment tier info

Amazon Elastic Beanstalk has two types of environment tiers to support different types of web applications.

☒ **Web server environment**
Run a website, web application, or web API that serves HTTP requests. [Learn more](#)

☐ **Worker environment**
Run a worker application that processes long-running workloads on demand or performs tasks on a schedule. [Learn more](#)

Application information info

Application name

Maximum length of 100 characters.

► **Application tags (optional)**

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Environment information info

Choose the name, subdomain and description for your environment. These cannot be changed later.

Environment name

Must be from 4 to 40 characters in length. The name can contain only letters, numbers, and hyphens. It can't start or end with a hyphen. This name must be unique within a region in your account.

Domain

Environment description

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Platform info

Platform type

☒ **Managed platform**
Platforms published and maintained by Amazon Elastic Beanstalk. [Learn more](#)

☐ **Custom platform**
Platforms created and owned by you. This option is unavailable if you have no platforms.

Platform

Platform branch

Platform version

Application code info

☒ **Sample application**
☐ Existing version
Application versions that you have uploaded.

☐ **Upload your code**
Upload a source bundle from your computer or copy one from Amazon S3.

Application code info

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☐ Existing version
Application versions that you have uploaded.

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Upload a source bundle from your computer or copy one from Amazon S3.

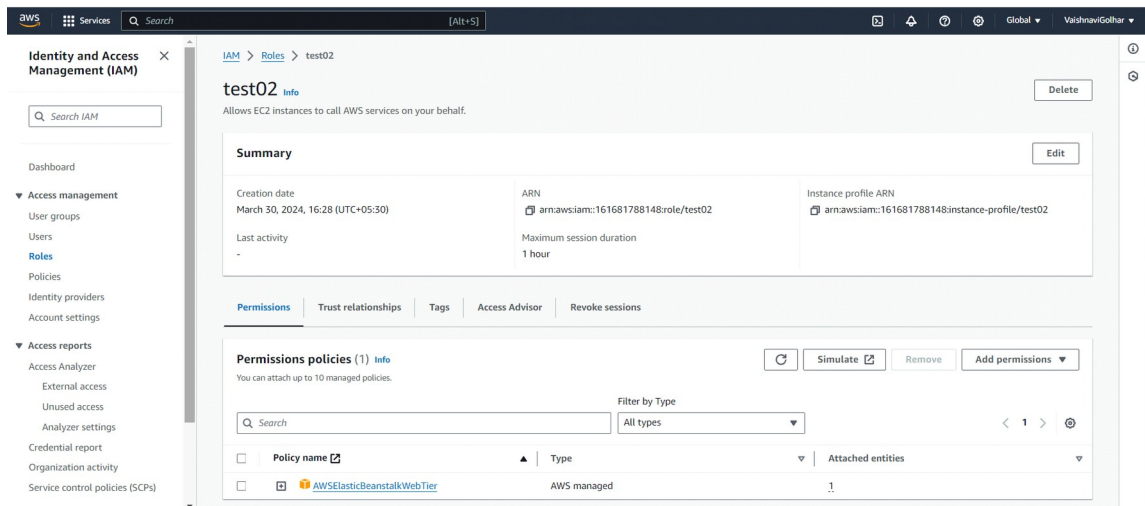
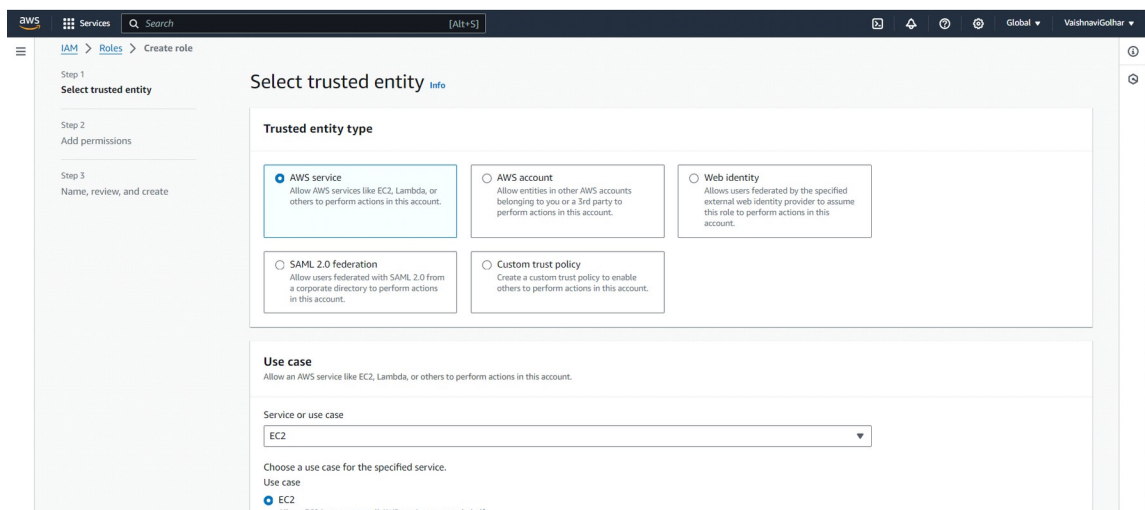
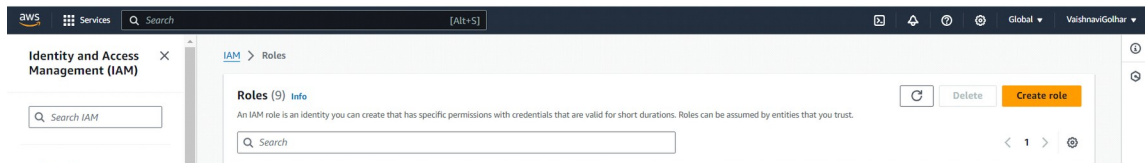
Presets info

Start from a preset that matches your use case or choose custom configuration to unset recommended values and use the service's default values.

Configuration presets
☒ **Single instance (free tier eligible)**
☐ Single instance (using spot instance)
☐ High availability
☐ High availability (using spot and on-demand instances)
☐ Custom configuration

Cancel

Next



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Step 1
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Configure service access info

Service access
IAM roles, assumed by Elastic Beanstalk as a service role, and EC2 instance profiles allow Elastic Beanstalk to create and manage your environment. Both the IAM role and instance profile must be attached to IAM managed policies that contain the required permissions. [Learn more](#)

Service role
☒ Create and use new service role
☐ Use an existing service role

Service role name
Enter the name for an IAM role that Elastic Beanstalk will create to assume as a service role. Beanstalk will attach the required managed policies to it.

[View permission details](#)

EC2 key pair
Select an EC2 key pair to securely log in to your EC2 instances. [Learn more](#)

[Create new key pair](#)

EC2 instance profile
Choose an IAM instance profile with managed policies that allow your EC2 instances to perform required operations.

[View permission details](#)

[Cancel](#) [Skip to review](#) [Previous](#) [Next](#)

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Step 1
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Set up networking, database, and tags - optional info

Virtual Private Cloud (VPC)
VPC
Launch your environment in a custom VPC instead of the default VPC. You can create a VPC and subnets in the VPC management console. [Learn more](#)

[Create custom VPC](#)

Instance settings
Choose a subnet in each AZ for the instances that run your application. To avoid exposing your instances to the Internet, run your instances in private subnets and load balancer in public subnets. To run your load balancer and instances in the same public subnets, assign public IP addresses to the instances. [Learn more](#)

Public IP address
Assign a public IP address to the Amazon EC2 instances in your environment.
☒ Activated

Instance subnets

☒	Availability Zone	Subnet	CIDR	Name
☒	ap-south-1c	subnet-0016caa4a...	172.31.16.0/20	
☒	ap-south-1b	subnet-00244b72...	172.31.0.0/20	
☒	ap-south-1a	subnet-0d3a60cdd...	172.31.32.0/20	

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Database info

Integrate an RDS SQL database with your environment. [Learn more](#)

Database subnets
If your Elastic Beanstalk environment is attached to an Amazon RDS, choose subnets for your database instances. [Learn more](#)

Choose database subnets (3)

☒	Availability Zone	Subnet	CIDR	Name
☒	ap-south-1c	subnet-0016caa4a...	172.31.16.0/20	
☒	ap-south-1b	subnet-00244b72...	172.31.0.0/20	
☒	ap-south-1a	subnet-0d3a60cdd...	172.31.32.0/20	

☐ Enable database

Restore a snapshot - optional
Restore an existing snapshot from a previously used database.
Snapshot

Database settings
Choose an engine and instance type for your environment's database.
Engine

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Database

Restore database

Restore a snapshot - optional

Restores an existing snapshot from a previously used database.

Snapshot

None

Database settings

Choose an engine and instance type for your environment's database.

Engine

Engine version

Instance class

Storage

Choose a number between 5 GB and 16384 GB.

16

GB

Username

Password

Availability

Use Oracle A-Z

Database deletion policy

Throttles apply when you drop a database or terminate the environment snapshot.

Create snapshot

Enables Amazon to create a snapshot of the database and then delete it. You can restore a database from a snapshot when you add a DB to an Amazon RDS instance or when you create a database instance. You might want to create a snapshot for testing database snapshots.

Backup

The Amazon database will remain available and operational external to Amazon RDS.

Delete

Enables Amazon to terminate the database. The database will no longer be available.

Tags

Apply up to 50 tags. You can use tags to group and filter your resources. A tag is a key-value pair. The key must be unique within the resource and is case sensitive. Learn more.

No tags associated with this resource.

Add new tag

You can add 50 more tags.

Cancel Skip to review Previous Next

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Database

Restore database

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Configure instance traffic and scaling - optional [Info](#)

▼ Instances [Info](#)

Configure the Amazon EC2 instances that run your application.

Root volume (boot device)

Root volume type
(Container default)

Size
The number of gigabytes of the root volume attached to each instance.
8 GB

IOPS
Input/output operations per second for a provisioned IOPS (SSD) volume.
100 IOPS

Throughput
The desired throughput to provision for the Amazon EBS root volume attached to your environment's EC2 instance
125 MIB/s

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Amazon CloudWatch monitoring

The time interval between when metrics are reported from the EC2 instances

Monitoring interval
5 minute

Instance metadata service (IMDS)

Your environment's platform supports both IMDSv1 and IMDSv2. To enforce IMDSv2, deactivate IMDSv1. [Learn more](#)

IMDSv1

With the current setting, the environment enables only IMDSv2.

☒ Deactivated

EC2 security groups

Select security groups to control traffic.

EC2 security groups (2)

☐	Group name	Group ID	Name
☐	default	sg-078c5922595f70261	
☒	RDS database sg	sg-07249788f98d0c84c	

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▼ Capacity [Info](#)

Configure the compute capacity of your environment and auto scaling settings to optimize the number of instances used.

Auto scaling group

Environment type

Select a single-instance or load-balanced environment. You can develop and test an application in a single-instance environment to save costs and then upgrade to a load-balanced environment when the application is ready for production. [Learn more](#)

Single instance

Instances

1 Min

1 Max

Fleet composition

Spot instances are launched at the lowest available price. [Learn more](#)

☒ On-Demand instance

☐ Spot instance

Maximum spot price

The maximum price per instance-hour, in USD, that you're willing to pay for a Spot Instance. Setting a custom price limits your chances to fulfill your target capacity using Spot instances.

☐ Default

☒ Set your maximum price

On-Demand base

The minimum number of On-Demand Instances that your Auto Scaling group provisions before considering Spot Instances as your environment scales out.

Configure your environment capacity using spot instances

☐ Default

☐ Set your maximum price

On-Demand base
The minimum number of On-Demand Instances that your Auto Scaling group provisions before considering Spot Instances as your environment scales out.

0

On-Demand above base
The percentage of On-Demand Instances as part of any additional capacity that your Auto Scaling group provisions beyond the On-Demand base instances.

0 %

Capacity rebalancing
Specifies whether to enable the capacity rebalancing feature for Spot Instances in your Auto Scaling Group. This option is only relevant when `EnabledSpot` is true in the `aws:ec2:instances` namespace, and there is at least one Spot Instance in your Auto Scaling group.

☐ Turn on capacity rebalancing

Architecture
The processor architecture determines the instance types that are made available. You can't change this selection after you create the environment. [Learn more](#)

☒ **x86_64**
This architecture uses x86 processors and is compatible with most third-party tools and libraries.

☐ **arm64 - new**
This architecture uses AWS Graviton2 processors. You might have to recompile some third-party tools and libraries.

Instance types
Add instance types for your fleet. Change the order that the instances are in to set the preferred launch order. This only affects On-Demand instances. We recommend you include at least two instance types. [Learn more](#)

Choose x86 instance types

t3.micro X t3.small X

☒ **x86_64**
This architecture uses x86 processors and is compatible with most third-party tools and libraries.

☐ **arm64 - new**
This architecture uses AWS Graviton2 processors. You might have to recompile some third-party tools and libraries.

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Add instance types for your fleet. Change the order that the instances are in to set the preferred launch order. This only affects On-Demand instances. We recommend you include at least two instance types. [Learn more](#)

Choose x86 instance types

t3.micro X t3.small X

AMI ID
Elastic Beanstalk selects a default Amazon Machine Image (AMI) for your environment based on the Region, platform version, and processor architecture that you choose. [Learn more](#)

ami-09a8a07a7c72990e4

Availability Zones
Number of Availability Zones (AZs) to use.

Any

Placement
Specify Availability Zones (AZs) to use.

Choose Availability Zones (AZs)

Scaling cooldown

360 seconds

Cancel Skip to review Previous Next

Elastic Beanstalk

Applications

Environments

Change history

Recent environments

Project2-env

MyPHPApp-env

Elastic Beanstalk is launching your environment. This will take a few minutes.

Elastic Beanstalk is launching your environment. This will take a few minutes.

Elastic Beanstalk > Environments

Environments (1) Info

Filter environments

Environment name	Health	Application	Platform	Domain	Running version	Tier name	Date created
Project2-env	Ok	project2	PHP 8.2 runni...	Project2-env.eba-tcmidip.ap-s...	-	WebServer	March 3

Environment creation takes about 5 minutes and creates the following resources:

EC2 instance – An Amazon Elastic Compute Cloud (Amazon EC2) virtual machine configured to run web apps on the platform that you choose.

Each platform runs a specific set of software, configuration files, and scripts to support a specific language version, framework, web container, or combination of these. Most platforms use either Apache or NGINX as a reverse proxy that sits in front of your web app, forwards requests to it, serves static assets, and generates access and error logs.

Instance security group – An Amazon EC2 security group configured to allow inbound traffic on port 80. This resource lets HTTP traffic from the load balancer reach the EC2 instance running your web app. By default, traffic isn't allowed on other ports.

Load balancer – An Elastic Load Balancing load balancer configured to distribute requests to the instances running your application. A load balancer also eliminates the need to expose your instances directly to the internet.

Load balancer security group – An Amazon EC2 security group configured to allow inbound traffic on port 80. This resource lets HTTP traffic from the internet reach the load balancer. By default, traffic isn't allowed on other ports.

Auto Scaling group – An Auto Scaling group configured to replace an instance if it is terminated or becomes unavailable.

Amazon S3 bucket – A storage location for your source code, logs, and other artifacts that are created when you use Elastic Beanstalk.

Amazon CloudWatch alarms – Two CloudWatch alarms that monitor the load on the instances in your environment and that are triggered if the load is too high or too low. When an alarm is triggered, your Auto Scaling group scales up or down in response.

AWS CloudFormation stack – Elastic Beanstalk uses AWS CloudFormation to launch the

resources in your environment and propagate configuration changes. The resources are defined in a template that you can view in the AWS CloudFormation console.

Domain name – A domain name that routes to your web app in the form `subdomain.region.elasticbeanstalk.com`.

All of these resources are managed by Elastic Beanstalk. When you terminate your environment, Elastic Beanstalk terminates all the resources that it contains. The RDS DB instance that you launched is outside of your environment, so you are responsible for managing its lifecycle.

3. Configure security groups, environment properties, and scaling

Add the security group of your DB instance to your running environment. This procedure causes Elastic Beanstalk to reprovision all instances in your environment with the additional security group attached.

a. To add a security group to your environment

1. To add a security group using the Elastic Beanstalk console. Open the Elastic Beanstalk console, and in the Regions list, select your AWS Region.

In the navigation pane, choose Environments, and then choose the name of your environment from the list.

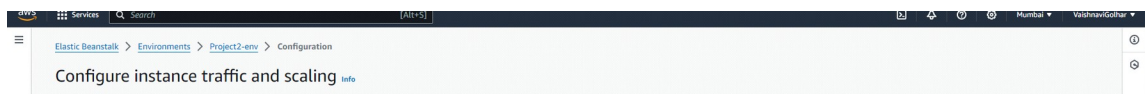
In the navigation pane, choose Configuration.

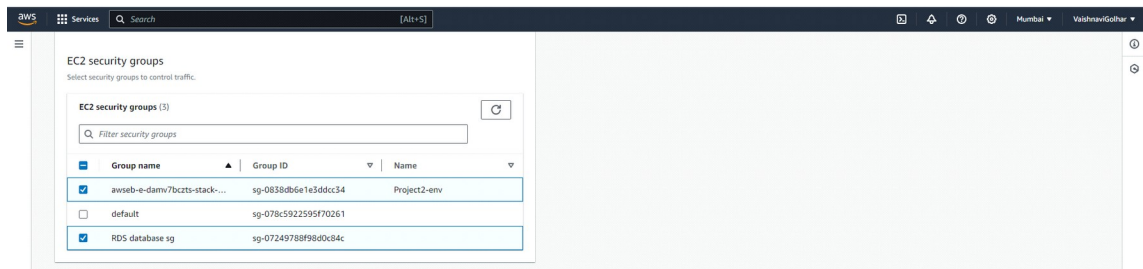
2. In the Instances configuration category, choose Edit.

Under EC2 security groups, choose the security group to attach to the instances, in addition to the instance security group that Elastic Beanstalk creates.

To save the changes choose Apply at the bottom of the page.

Read the warning, and then choose Confirm.





b. To configure environment properties for an Amazon RDS DB instance

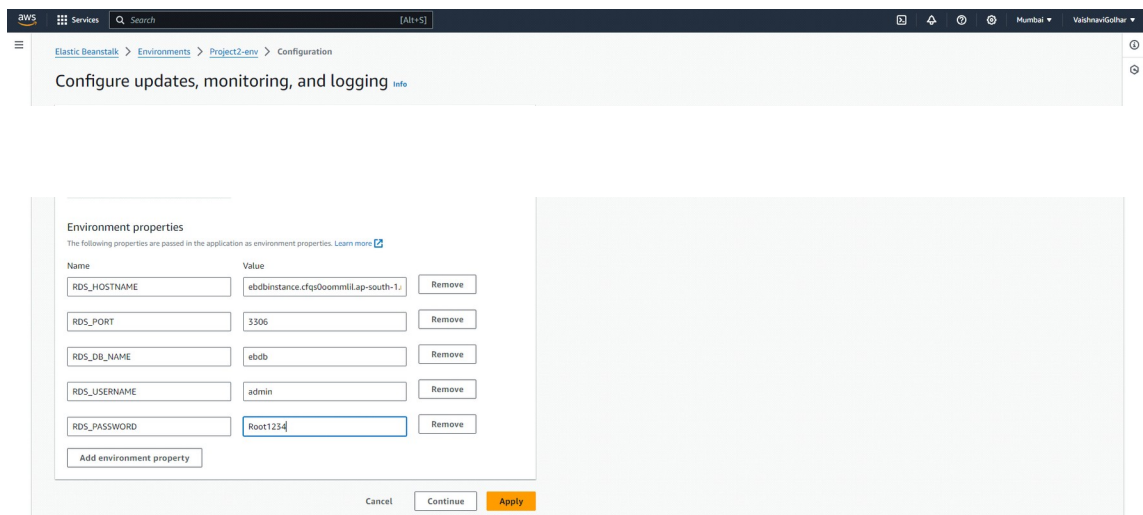
1. Open the Elastic Beanstalk console, and in the Regions list, select your AWS Region.

In the navigation pane, choose Environments, and then choose the name of your environment from the list.

2. In the navigation pane, choose Configuration.

3. In the Updates, monitoring, and logging configuration category, choose Edit.

In the Environment properties section, define the variables that your application reads to construct a connection string. For compatibility with environments that have an integrated RDS DB instance, use the following names and values. You can find all values, except for your password, in the RDS console.

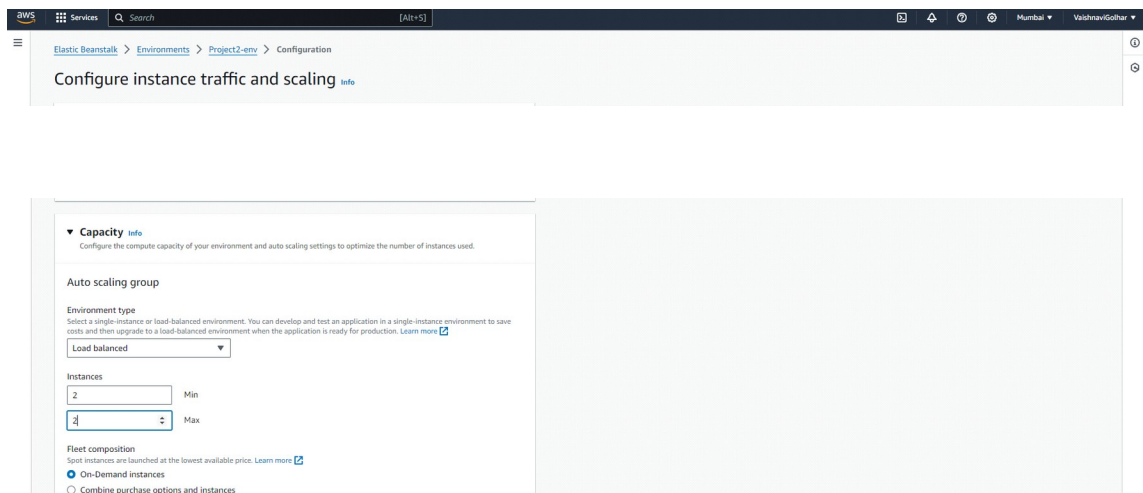


c. To configure your environment's Auto Scaling group for high availability

1. Open the Elastic Beanstalk console, and in the Regions list, select your AWS Region.

In the navigation pane, choose Environments, and then choose the name of your environment from the list.

2. In the navigation pane, choose Configuration. In the Capacity configuration category, choose Edit. In the Auto Scaling group section, set Min instances to 2. To save the changes choose Apply at the bottom of the page.

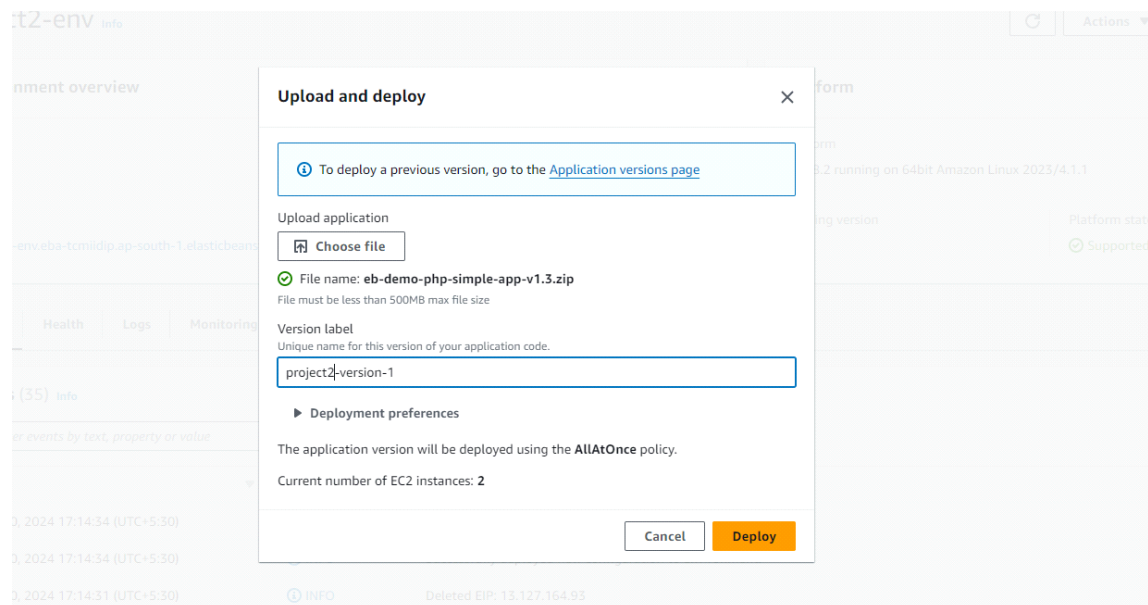


d. To deploy a source bundle

1. On the environment overview page, choose Upload and deploy.

Use the on-screen dialog box to upload the source bundle.

Choose Deploy.



When the deployment completes, you can choose the site URL to open your website in a new tab.

The site collects user comments and uses a MySQL database to store the data. To add a comment, choose Share Your Thought, enter a comment, and then choose Submit Your Thought. The web app writes the comment to the database so that any instance in the environment can read it, and it won't be lost if instances go out of service.



